

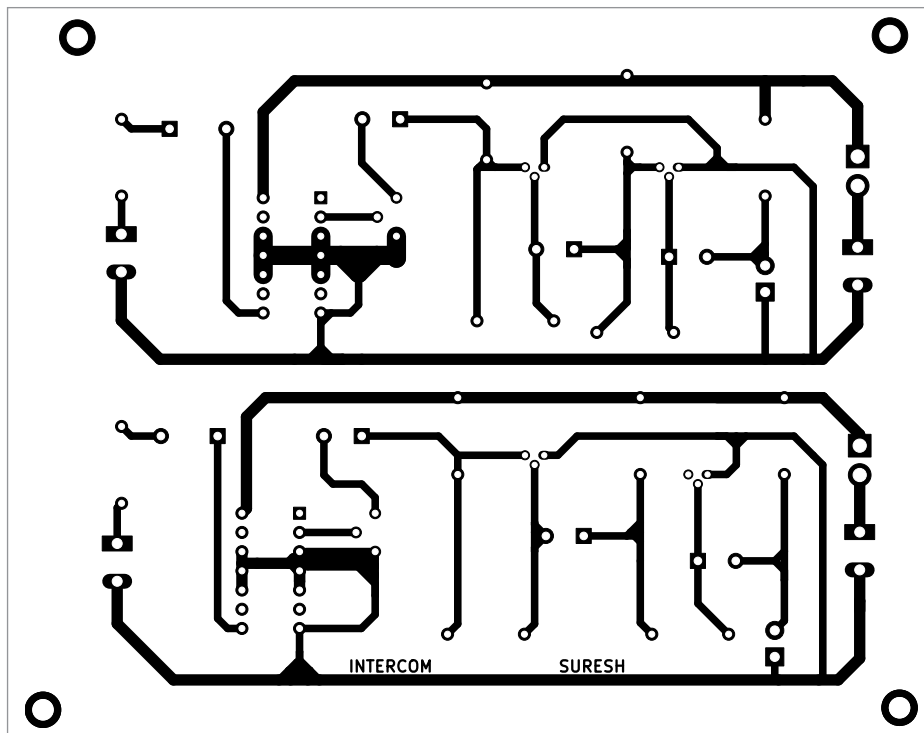


Fig. 3: Actual-size PCB layout

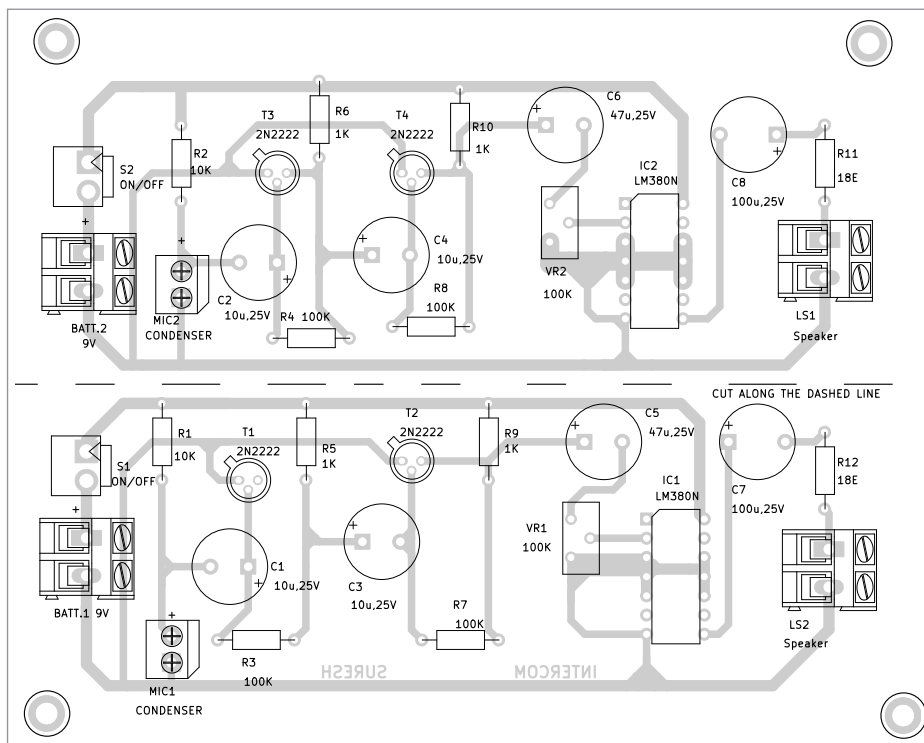


Fig. 4: Component layout of the PCB

LM380N via capacitor C7 and resistor R12 reproduces the sound. A 9V regulated supply or 9V battery each is required to power the upper and

lower sections of the circuit.

The upper section of the circuit is identical and works the same way. When you speak in front of

microphone MIC1, your voice can be heard from speaker LS2. And when you speak in front of microphone MIC2, your voice can be heard from speaker LS1. Thus, 2-way communication becomes possible.

Construction and testing

An actual-size PCB layout for the intercom is shown in Fig. 3 and its component layout in Fig. 4.

Assemble the circuit on the PCB or a general-purpose board. Cut the PCB into two parts (SET1 and SET2) along the dotted line marked on it. Enclose each PCB separately in a suitable box. Place SET1 in one room and SET2 in another. Connect LS2 in SET1 in first room to SET2 in the second room and LS1 from SET2 to SET1 in first room using a twin wire.

Fix MIC1 on the front side and LS2 on the rear side of the SET1 enclosure. Similarly, fix MIC2 on the front side and LS1 on the rear side of SET2 enclosure. Switch on S1 in SET1 and S2 in SET2 to start communication from both sides.

When you speak into microphone MIC1 in SET1, the person in the second room will hear you through loudspeaker LS2. Similarly, when he speaks into microphone MIC2, you will hear him through loudspeaker LS1.

Please note, there will be acoustical feedback between the microphone and loudspeaker if they are close and facing each other. So, keep distance between the microphones and loudspeakers while assembling and fixing them on the enclosures.

EFY

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